Firm's Production Function

Pascal Michaillat
https://pascalmichaillat.org/w/

Firms employ 2 types of workers:

- $l - n$ recruiters → fill vacant jobs
  $\hat{\tau}(\theta) \times n$ → do not participate in production
  → HR department

- n producers → participate in production

Production function:

$$\boxed{k = a \times n^{\alpha}}$$

- $a > 0$: technology of firm
- $\alpha \in (0, 1)$: diminishing marginal returns to labor
  ↳ required to obtain nondegenerate labor demand (Michaillat 2012)
- $n > 0$: # producers (not # employees)
- $k > 0$: firm's capacity → # services that could be sold w/ enough customers
  → in practice firms will sell less than capacity b/c of matching function
- what is sold is $\boxed{f(x) \cdot k < k}$
  selling proba